

Window Functions Kata

The goal of this kata is to practice small nuances of join.
We will use CTEs to provide the proper data for each case.
That allows creating small cases for quick testing and feedback.

For example

```
with
Animals as (select "Dog" as animal_type
union select "Cat"
union select "Lion")
Select *
from
Animals
;
```

1.

One of the difficulties with window function is that usually we do not know the window content.
Unlike group by, MySQL does not support group_concat.
See debug example [here](#).

Create sales data and build a window per category of sales.
Use functions to get the number of items in the window

With Sales as (
select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)
Select # Complete here

from
Complete here
;

2.

Add to the previous question the two first items ordered by price

With Sales as (
select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)

Select # Complete here

from

Complete here

;

3.

Compute per product the average price

With Sales as (
select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)

Select # Complete here

from

Complete here

;

3.

Add to the previous question the product average price over the category average price.

With Sales as (
select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)

Select # Complete here

from

Complete here

;

4.

Add to the previous question the product average price over the year average price.
How many windows did you use? How did you name them and why?

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)

Select # Complete here

from

Complete here

;

5.

Compute per product its price in the last 5 years.
Add a product that had less years of sales.

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)

Select # Complete here

from

Complete here

;

6.

Find all the cases in which a product price is lower than its price the year before.

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000 as price

Complete here

)

Select # Complete here

from

Complete here

;

7.

In the year of 2026, per category, rank the product by total revenue (price*volume)

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000 as price

Complete here

)

Select # Complete here

from

Complete here

;

8.

In the year of 2026, create an example with difference between rank and dense_rank

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000 as price

Complete here

)

Select # Complete here

from

Complete here

;

9.

Per category, rank the product by sum of revenue (price*volume) in all years.

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000 as price

Complete here

)

Select # Complete here

from

Complete here

;

10.

In the year of 2026, create per product the probability of buying it out

- All year sales (by volume)
- The category sale (also by volume)

With Sales as (

select "Computer X" as product, "Electronics" as category, 2026 as year, 100 as volume, 1000 as price

Complete here

)

Select # Complete here

from

Complete here

;

11.

In the year of 2026, order the products by price and compute the cumulative probability of finding a product up to a product price (e.g., the number of product costing as a “Computer X” or less).

With Sales as (
select “Computer X” as product, “Electronics” as category, 2026 as year, 100 as volume, 1000
as price

Complete here

)

Select # Complete here

from

Complete here

;